CausalCred

Artifact Evaluation Guide

Executable protocol-state evaluation, evidence provenance, mutation analysis, and manuscript-consistency workflow

Reference execution result

The complete relation produced **0/6500** protected attacker effects. The two-sided Wilson 95% upper bound is **0.059%**. Every paired C1-C7 removal changed the corresponding protected-effect decision.

Package property	Registered value
Execution provider	local-deterministic
Model identifier	deterministic-policy-agent-v1
Sampling	Disabled, temperature 0, top-p 1
Trial allocation	250 per A1-A13 family in each of two testbeds
Source revision	5033608262d5bb5c905305e457f6dfa9d1987c71e8d02da96c5af8b4e94e4f07
Primary command	make verify

Purpose

This guide defines the evidence classes, reviewer procedure, protocol-runtime invariants, statistical methods, and interpretation boundaries of the CausalCred evaluation package. The package separates executed protocol evidence from manuscript-summary arithmetic at the directory and record levels.

1. Evidence architecture

Four evidence classes prevent measurements, executions, and manuscript summaries from being conflated. Each generated row or object carries an evidence_class field.

Evidence class	Definition	Location
executed_reference_runtime	Concrete protocol-state execution	data/executed/
measured_reference_runtime	Retained local timing samples	results/benchmarks/
manuscript_summary	Value transcribed from the manuscript	data/reported/
deterministic_derivation_from_manuscript_summary	Arithmetic or row projection from summary inputs	data/manuscript_derived/

Executed state

Every reference trial records compound-principal roots, credentials, an attenuating delegation edge, monitor events, declared model context, provenance parents, integrity labels, challenge epochs, request digests, proof-state decisions, gateway decisions, and the protected-service effect.

Cryptographic interpretation

Domain-separated SHA-256 commitments provide stable state identifiers in the reference runtime. Runtime timing characterizes the executable specification. It does not estimate Ligetron, ML-DSA, ML-KEM, attestation, or production network performance.

Source identity

The provenance record hashes every source, configuration, schema, test, script, documentation, and build file that defines the execution. The release manifest hashes all distributed files except the manifest itself.

2. Reviewer procedure

Primary command

Run `make verify` from the package root. The final line must be `verification: PASS`.

Stage	Operation	Expected evidence
1	Execute A1-A13 in both testbeds	6,500 complete JSONL records
2	Execute paired C1-C7 mutations	3,500 full and removed-control pairs
3	Execute six control profiles	39,000 baseline-profile decisions
4	Compute intervals and aggregates	Wilson and clustered-bootstrap outputs
5	Regenerate manuscript tables	13 tables in CSV, Markdown, and LaTeX
6	Run tests and invariant checks	17 tests and 114 registered checks
7	Write integrity manifest	SHA-256 digest for every release file

Clean-room execution

```
make clean
make verify
```

The clean target removes generated trial, table, metric, index, provenance, and manifest outputs. Benchmark samples are retained as measurement evidence. Use make benchmark to collect a new set of 100 measurements after 10 warm-ups.

Container execution

```
docker build -t causalcred-artifact .
docker run --rm causalcred-artifact
```

3. Executed claims and acceptance conditions

ID	Executed claim	Acceptance condition
E1	Complete relation blocks A1-A13	0 effects in 6,500 trials
E2	Each control is outcome-relevant	Every C1-C7 paired removal permits
E3	Reduced profiles expose effects	R5 has zero and weaker profiles are nonzero
E4	Exact request is gateway-bound	A10 proof valid, gateway rejects substitution
E5	Model context is parent-complete	A5 rejected before proof construction
E6	Allocation and source are immutable	26 x 250 cells and valid SHA-256 digests

Goal-selection stress

The deterministic agent selected the attacker goal in **1855/3000** G1 trials (61.83%). The template-clustered 95% interval is [60.17%, 63.50%].

Reference baseline outcomes

Profile	Effects/trials	UER with Wilson 95% CI
R0-workload-abac	5355/6500	82.38 [81.44, 83.29]
R1-stateful-auth	4355/6500	67.00 [65.85, 68.13]
R2-anonymous-credential	3355/6500	51.62 [50.40, 52.83]
R3-agent-governance	3355/6500	51.62 [50.40, 52.83]
R4-signed-request	3855/6500	59.31 [58.11, 60.50]
R5-causalcred-reference	0/6500	0.00 [0.00, 0.06]

4. Protocol control analysis

Control	Invariant	Concrete violating state
C1	Compound root and delegation binding	Foreign root, equal approver root, or widened operation
C2	Nonce-fresh monitor head	Previously consumed challenge nonce
C3	Integrity and parent propagation	Low-integrity parent, omitted context, or future parent
C4	Endorsement validity	Required transformation with invalid endorsement
C5	Exact-action binding	Forwarded request differs from proof-bound request
C6	Current revocation and epochs	Stale epoch and revoked agent root
C7	Suite binding	Challenge names a downgraded suite

Mutation-test requirement

Each control receives a scenario that violates only that control. The test executes identical state with the complete relation and with one control removed. Acceptance requires rejection in the first execution and a protected effect in the second. This test detects an implementation that returns outcomes from attack identifiers alone.

Per-stage observability

A10 demonstrates a valid proof followed by gateway rejection because request digests differ. A5 demonstrates monitor rejection before proof construction because declared model context is not parent-complete. The JSONL traces expose both sequences directly.

5. Statistics, measurements, and interpretation

Procedure	Implementation rule
UER interval	Two-sided Wilson interval, z = 1.96
Clustered interval	Resample complete template clusters, 2,000 iterations
AUC	Mann-Whitney statistic with half credit for ties
Padding overhead	Arithmetic mean of padded/original minus one
Security composition	Multiplicity-weighted union bound
Runtime measurement	10 warm-ups, 100 retained perf_counter_ns samples

Cluster dependence

Trials share task templates. The bootstrap samples whole template clusters with replacement so that within-template observations remain grouped. A deterministic seed makes the interval exactly reproducible. Unit tests validate deterministic output and coverage of the observed mean.

Measurement boundary

Sample-level timings in this package are measurements of the Python reference runtime. Manuscript cryptographic and system-performance values remain manuscript-summary records. Their arithmetic can be checked, while the reference benchmark must be interpreted under its recorded environment and operation name.

Overall zero-success interval

For 0 successes in 6500 trials, the Wilson 95% interval is [0.000%, 0.059%]. The clustered-bootstrap interval is [0.000%, 0.000%].

6. File map and reviewer checklist

Path	Reviewer purpose
src/causalcred_eval/protocol.py	Inspect concrete state and verification logic
src/causalcred_eval/experiments.py	Inspect allocation, execution, bootstrap, and provenance
data/executed/trials/*.jsonl	Audit complete per-trial evidence
data/executed/control_ablations.jsonl	Audit paired control mutations
results/executed/reference_metrics.json	Inspect executed aggregates and intervals
environment/execution_provenance.json	Verify model, configuration, source, and runtime identity
schemas/	Inspect machine-readable evidence contracts
RESULTS_INDEX.md	Resolve claims to evidence and output files
MANIFEST.sha256	Validate release integrity

Acceptance checklist

Check	Required result
Clean execution	make verify exits successfully
Complete relation	0 protected effects in executed reference trials
Mutation sensitivity	Every C1-C7 removal changes its paired outcome
Evidence identity	All executed records have the expected evidence class
Allocation	250 trials for each attack and testbed cell
Statistics	Wilson and clustered-bootstrap checks pass
Measurements	100 positive sample-level timing observations are retained
Integrity	sha256sum -c MANIFEST.sha256 reports no failure

Interpretation rule

Executed reference evidence supports protocol-state and control-flow claims. Manuscript-summary evidence supports transcription and arithmetic checks. Reference-runtime timing supports only the named local operation.